# Flow matching and Diffusion Models

## Lecture 3: Training the Generative Model

### flow matching

flow model given by: $X_0 \sim p_{init}$, $dX_t = u_t^\theta(X_t)dt$

neural network $u_t^\theta$ ($\theta$ parameters)

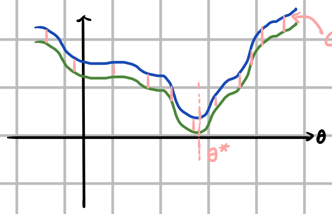
- Goal: $u_t^\theta \approx u_t^{target}$
- Flow matching loss: $\mathcal{L}_{FM}(\theta) = \mathbb{E}_{t\sim Unif,\, x\sim p_t}[\|u_t^\theta(x) - u_t^{target}(x)\|^2] = \mathbb{E}_{t\sim Unif,\, z\sim p_{data},\, x\sim p_t(\cdot|z)}[\|u_t^\theta(x) - u_t^{target}(x)\|^2]$

uniform in $[0,1]$ $\quad u_t^{target}(x) = \int u_t^{target}(x|z)\frac{p_t(x|z)p_{data}(z)}{p_t(x)}dz \Leftarrow$ intractable

conditional flow matching loss:

$\mathcal{L}_{CFM}(\theta) = \mathbb{E}_{t\sim unif,\, z\sim p_{data},\, x\sim p_t(\cdot|z)}[\|u_t^\theta(x) - u_t^{target}(x|z)\|^2] \Leftarrow$ tractable

$\mathcal{L}_{FM}(\theta) = \mathcal{L}_{CFM}(\theta) + C \Rightarrow C$ is independent of $\theta$, $C>0$




① for minimizer $\theta^*$ of $\mathcal{L}_{CFM}$: $u_t^{\theta^*} = u_t^{target}$

② $\nabla_\theta \mathcal{L}_{FM}(\theta) = \nabla_\theta \mathcal{L}_{CFM}(\theta)$ $\qquad \nabla_\theta \mathcal{L}(\theta) = [\frac{\partial \mathcal{L}}{\partial \theta_1}, \frac{\partial \mathcal{L}}{\partial \theta_2}, \cdots \frac{\partial \mathcal{L}}{\partial \theta_n}]$

SGD (stochastic gradient descent) the same

### example: $\mathcal{L}_{CFM}$ for Gaussian conditional path.

$p_t(\cdot|z) = \mathcal{N}(\alpha_t z, \beta_t^2 I_d)$ $\quad t: 0\to 1$ $\quad \alpha_t: 0\to 1$ $\quad \beta_t: 1\to 0$

$u_t^{target}(x|z) = (\dot\alpha_t - \frac{\dot\beta_t}{\beta_t}\alpha_t) z + \frac{\dot\beta_t}{\beta_t} x$

$\epsilon \sim \mathcal{N}(0, I_d) \Rightarrow x_t = \alpha_t z + \beta_t \epsilon \sim \mathcal{N}(\alpha_t z, \beta_t^2 I_d) = p_t(\cdot|z)$

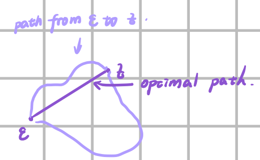
$\mathcal{L}_{CFM}(\theta) = \mathbb{E}_{t\sim unif,\, z\sim p_{data},\, x\sim \mathcal{N}(\alpha_t z, \beta_t^2 I_d)}[\|u_t^\theta(x) - (\dot\alpha_t - \frac{\dot\beta_t}{\beta_t}\alpha_t) z - \frac{\dot\beta_t}{\beta_t} x\|^2]$

$x = \alpha_t z + \beta_t \epsilon$

$(\dot\alpha_t - \frac{\dot\beta_t}{\beta_t}\alpha_t) z + \frac{\dot\beta_t}{\beta_t}(\alpha_t z + \beta_t \epsilon) = \dot\alpha_t z + \dot\beta_t \epsilon$

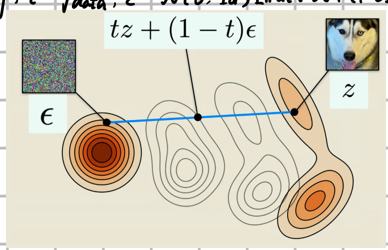
$= \mathbb{E}_{t\sim unif,\, z\sim p_{data},\, \epsilon\sim \mathcal{N}(0, I_d)}[\|u_t^\theta(\alpha_t z + \beta_t \epsilon) - (\dot\alpha_t z + \dot\beta_t \epsilon)\|^2]$

### CondOT (Conditional Optimal Transport) probability path.




$\alpha_t = t \Rightarrow \dot\alpha_t = 1$ $\quad \beta_t = 1-t$ $\quad \dot\beta_t = -1$

$= \mathbb{E}_{t\sim unif,\, z\sim p_{data},\, \epsilon\sim \mathcal{N}(0, I_d)}[\|u_t^\theta(tz + (1-t)\epsilon) - (z-\epsilon)\|^2]$




**Algorithm 3** Flow Matching Training Procedure (here for Gaussian CondOT path $p_t(x|z) = \mathcal{N}(tz, (1-t)^2)$)

**Require:** A dataset of samples $z \sim p_{\text{data}}$, neural network $u_t^\theta$

1: **for** each mini-batch of data **do**
2: Sample a data example $z$ from the dataset.
3: Sample a random time $t \sim \text{Unif}_{[0,1]}$.
4: Sample noise $\epsilon \sim \mathcal{N}(0, I_d)$
5: Set $x = tz + (1-t)\epsilon$ (General case: $x \sim p_t(\cdot|z)$)
6: Compute loss

$$\mathcal{L}(\theta) = \|u_t^\theta(x) - (z-\epsilon)\|^2 \qquad \text{(General case: } = \|u_t^\theta(x) - u_t^{\text{target}}(x|z)\|^2)$$

7: Update the model parameters $\theta$ via gradient descent on $\mathcal{L}(\theta)$.
8: **end for**

proof: $\mathcal{L}_{FM}(\theta) = \mathcal{L}_{CFM}(\theta) + C$:

$\|a-b\|^2 = \|a\|^2 - 2a^Tb + \|b\|^2$ $\quad (a, b \in \mathbb{R}^d)$

$\mathcal{L}_{FM}(\theta) = \mathbb{E}_{t\sim Unif,\, x\sim p_t}[\|u_t^\theta(x) - u_t^{target}(x)\|^2]$

$= \mathbb{E}_{t\sim Unif,\, x\sim p_t}[\|u_t^\theta(x)\|^2 - 2u_t^\theta(x)^T u_t^{target}(x) + \|u_t^{target}(x)\|^2]$

$= \mathbb{E}_{t\sim Unif,\, x\sim p_t}[\|u_t^\theta(x)\|^2] - 2\mathbb{E}_{t\sim Unif,\, x\sim p_t}[u_t^\theta(x)^T u_t^{target}(x)] + \mathbb{E}_{t\sim Unif,\, x\sim p_t}[\|u_t^{target}(x)\|^2] = C_1$

$= \mathbb{E}_{t\sim Unif,\, z\sim p_{data},\, x\sim p_t(\cdot|z)}[\|u_t^\theta(x)\|^2] - 2\mathbb{E}_{t\sim Unif,\, x\sim p_t}[u_t^\theta(x)^T u_t^{target}(x)] + C_1$

$E_{t,x}[g(t,x)] = \iint (\text{probability density})(\text{function value})\,dx\,dt.$

$\mathbb{E}_{t\sim Unif,\, x\sim p_t}[u_t^\theta(x)^T u_t^{target}(x)] = \int_0^1\int (1\cdot p_t(x))(u_t^\theta(x)^T u_t^{target}(x))\,dx\,dt$

$u_t^{target}(x) = \int u_t^{target}(x|z)\frac{p_t(x|z)p_{data}(z)}{p_t(x)}dz$

$= \iiint \cancel{p_t(x)}\, u_t^\theta(x)^T u_t^{target}(x|z)\frac{p_t(x|z)p_{data}(z)}{\cancel{p_t(x)}}\,dx\,dz\,dt = \iiint (p_t(x|z)p_{data}(z))(u_t^\theta(x)^T u_t^{target}(x|z))\,dx\,dz\,dt.$

$= \mathbb{E}_{t\sim Unif,\, z\sim p_{data},\, x\sim p_t(\cdot|z)}[u_t^\theta(x)^T u_t^{target}(x|z)]$

$= \mathbb{E}_{t\sim Unif,\, z\sim p_{data},\, x\sim p_t(\cdot|z)}[\|u_t^\theta(x)\|^2] - 2\mathbb{E}_{t\sim Unif,\, z\sim p_{data},\, x\sim p_t(\cdot|z)}[u_t^\theta(x)^T u_t^{target}(x|z)] + C_1$

$= \mathbb{E}_{t\sim Unif,\, z\sim p_{data},\, x\sim p_t(\cdot|z)}[\|u_t^\theta(x)\|^2 - 2u_t^\theta(x)^T u_t^{target}(x|z) + \|u_t^{target}(x|z)\|^2 - \|u_t^{target}(x|z)\|^2] + C_1$

$= \mathbb{E}_{t\sim Unif,\, z\sim p_{data},\, x\sim p_t(\cdot|z)}[\|u_t^\theta(x) - u_t^{target}(x|z)\|^2] + \underbrace{\mathbb{E}_{t\sim Unif,\, z\sim p_{data},\, x\sim p_t(\cdot|z)}[-\|u_t^{target}(x|z)\|^2]}_{C_2} + C_1$

$= \mathcal{L}_{CFM}(\theta) + C_2 + C_1 = \mathcal{L}_{CFM}(\theta) + C$

Score matching

marginal score function: $\nabla \log p_t(x) = \int \nabla \log p_t(x|z)\frac{p_t(x|z)p_{data}(z)}{p_t(x)}dz$

$X_0 \sim p_{init},\quad dX_t = [u_t^{target}(X_t) + \frac{\sigma_t^2}{2}\nabla \log p_t(X_t)]\,dt + \sigma_t\, dW_t \Rightarrow X_t \sim p_t$

score network: $s_t^\theta$ ($\theta$ parameters)

- Goal: $s_t^\theta \approx \nabla \log p_t(x)$
- Score matching loss (Denoising score match loss):

$\mathcal{L}_{SM}(\theta) = \mathbb{E}_{t\sim Unif,\, z\sim p_{data},\, x\sim p_t(\cdot|z)}[\|s_t^\theta(x) - \nabla \log p_t(x)\|^2]$ → intractable

conditional score matching loss:

$\mathcal{L}_{CSM}(\theta) = \mathbb{E}_{t\sim Unif,\, z\sim p_{data},\, x\sim p_t(\cdot|z)}[\|s_t^\theta(x) - \nabla \log p_t(x|z)\|^2]$ → tractable

$\mathcal{L}_{SM}(\theta) = \mathcal{L}_{CSM}(\theta) + C$ → for $C<0$ of $\theta$

① for minimizer $\theta^*$ of $\mathcal{L}_{CSM}$: $s_t^{\theta^*} = \nabla \log p_t$

② $\nabla_\theta \mathcal{L}_{SM}(\theta) = \nabla_\theta \mathcal{L}_{CSM}(\theta)$

Example: Denoising score matching for Gaussian probability path.

$\nabla \log p_t(x|z) = -\frac{x - \alpha_t z}{\beta_t^2}$

$\epsilon \sim N(0, I_d) \qquad x = \alpha_t z + \beta_t \epsilon \sim N(\alpha_t z, \beta_t^2 I_d)$

$\mathcal{L}_{CSM}(\theta) = \mathbb{E}_{t\sim Unif,\, z\sim p_{data},\, x\sim p_t(\cdot|z)}[\|s_t^\theta(x) + \frac{x-\alpha_t z}{\beta_t^2}\|^2]$

$x = \alpha_t z + \beta_t \epsilon.$

$= \frac{\alpha_t z + \beta_t\epsilon - \alpha_t z}{\beta_t^2} = \frac{\beta_t \epsilon}{\beta_t^2} = \frac{\epsilon}{\beta_t}$

$= \mathbb{E}_{t\sim Unif,\, z\sim p_{data},\, x\sim p_t(\cdot|z)}[\|s_t^\theta(\alpha_t z + \beta_t\epsilon) + \frac{\epsilon}{\beta_t}\|^2]$

$= \mathbb{E}_{t\sim Unif,\, z\sim p_{data},\, x\sim p_t(\cdot|z)}[\frac{1}{\beta_t^2}\|\beta_t s_t^\theta(\alpha_t z + \beta_t\epsilon) + \epsilon\|^2]$

**Algorithm 4** Score Matching Training Procedure for Gaussian probability path

**Require:** A dataset of samples $z \sim p_{\text{data}}$, score network $s_t^\theta$ or noise predictor $\epsilon_t^\theta$

**for** each mini-batch of data **do**
  Sample a data example $z$ from the dataset.
  Sample a random time $t \sim \text{Unif}_{[0,1]}$.
  Sample noise $\epsilon \sim \mathcal{N}(0, I_d)$
  Set $x_t = \alpha_t z + \beta_t \epsilon$ (General case: $x_t \sim p_t(\cdot|z)$)
  Compute loss

Numerically unstable for low $\beta$

$$\mathcal{L}(\theta) = \|s_t^\theta(x_t) + \frac{\epsilon}{\beta_t}\|^2 \qquad \text{(General case: } = \|s_t^\theta(x_t) - \nabla \log p_t(x_t|z)\|^2)$$

Alternatively: $\mathcal{L}(\theta) = \|\epsilon_t^\theta(x_t) - \epsilon\|^2$

  Update the model parameters $\theta$ via gradient descent on $\mathcal{L}(\theta)$.
**end for**

$\mathcal{L}_{CSM}(\theta) = \mathbb{E}_{t\sim Unif,\, z\sim p_{data},\, x\sim p_t(\cdot|z)}[\frac{1}{\beta_t^2}\|\beta_t s_t^\theta(x_t) + \epsilon\|^2]$

$s_t^\theta(x_t) = -\frac{\epsilon_t^\theta(x_t)}{\beta_t}$

$= \mathbb{E}_{t\sim Unif,\, z\sim p_{data},\, x\sim p_t(\cdot|z)}[\frac{1}{\beta_t^2}\|\epsilon - \epsilon_t^\theta(x_t)\|^2]$

$\Rightarrow \theta^*$ of $\mathcal{L}(\theta) = \| s_t^\theta(x_t) + \frac{\varepsilon}{\beta_t} \|^2$ equals to $\theta^*$ of $\mathcal{L}(\theta) = \| \varepsilon_t^\theta(x_t) - \varepsilon \|^2$

Stochastic sampling with diffussion models:

$$X_0 \sim p_{init} \quad dX_t = \left[u_t^{target}(X_t) + \frac{\sigma_t^2}{2} \nabla \log p_t(X_t)\right] dt + \sigma_t dW_t \Rightarrow X_t \sim p_t$$

plugin neural networks.

$$X_0 \sim p_{init} \quad dX_t = \left[u_t^\theta(X_t) + \frac{\sigma_t^2}{2} s_t^\theta(X_t)\right] dt + \sigma_t dW_t \xRightarrow{\text{after training.}} X_t \sim p_t$$

Denoising diffusion models = Diffusion models with a Gaussian probability path. (our standard example) $\mathcal{N}(\alpha_t z, \beta_t^2 I_d)$

Conversion formula for Gaussian probability path.

$$u_t^{target}(x|z) = \left(\beta_t^2 \frac{\dot\alpha_t}{\alpha_t} - \dot\beta_t \beta_t\right) \nabla \log p_t(x|z) + \frac{\dot\alpha_t}{\alpha_t} x$$

$$u_t^{target}(x) = \left(\beta_t^2 \frac{\dot\alpha_t}{\alpha_t} - \dot\beta_t \beta_t\right) \nabla \log p_t(x) + \frac{\dot\alpha_t}{\alpha_t} x$$

proof:

$$u_t^{target}(x|z) = \left(\dot\alpha_t - \frac{\dot\beta_t}{\beta_t}\alpha_t\right) z + \frac{\dot\beta_t}{\beta_t} x = \left(\beta_t^2 \frac{\dot\alpha_t}{\alpha_t} - \dot\beta_t \beta_t\right)\left(\frac{\alpha_t z - x}{\beta_t^2}\right) + \frac{\dot\alpha_t}{\alpha_t} x = \left(\beta_t^2 \frac{\dot\alpha_t}{\alpha_t} - \dot\beta_t \beta_t\right) \nabla \log p_t(x|z) + \frac{\dot\alpha_t}{\alpha_t} x$$

$$u_t^{target}(x) = \int u_t^{target}(x|z) \frac{p_t(x|z) p_{data}(z)}{p_t(x)} dz = \int \left[\left(\beta_t^2 \frac{\dot\alpha_t}{\alpha_t} - \dot\beta_t \beta_t\right) \nabla \log p_t(x|z) + \frac{\dot\alpha_t}{\alpha_t} x\right] \frac{p_t(x|z) p_{data}(z)}{p_t(x)} dz$$

$$= \left(\beta_t^2 \frac{\dot\alpha_t}{\alpha_t} - \dot\beta_t \beta_t\right) \int \nabla \log p_t(x|z) \frac{p_t(x|z) p_{data}(z)}{p_t(x)} dz + \frac{\dot\alpha_t}{\alpha_t} \int x \frac{p_t(x|z) p_{data}(z)}{p_t(x)} dz$$

$= \nabla \log p_t(x)$

Bayes' Theorem: $\frac{p_t(x|z) p_{data}(z)}{p_t(x)} = p_t(z|x)$

$$\int x \frac{p_t(x|z) p_{data}(z)}{p_t(x)} dz = x \int p_t(z|x) dz = x$$

$$= \left(\beta_t^2 \frac{\dot\alpha_t}{\alpha_t} - \dot\beta_t \beta_t\right) \nabla \log p_t(x) + \frac{\dot\alpha_t}{\alpha_t} x$$

$$\Rightarrow u_t^\theta = \left(\beta_t^2 \frac{\dot\alpha_t}{\alpha_t} - \dot\beta_t \beta_t\right) s_t^\theta(x) + \frac{\dot\alpha_t}{\alpha_t} x$$